Saturday 14 November | 11:30–12:00

Ternary semi-direct products in algebraically coherent categories

Arnaud Duvieusart

IRMP, UCLouvain - FNRS, Belgium

In the category of groups, an action of B on A can be defined equivalently as a function $B \times A \to A$ (satisfying certain axioms) or as a morphism $B \to \mathrm{Aut}(A)$. In particular, an action of a coproduct $B + C$ or a semi-direct product $B \rtimes C$ on A can easily be described by a pair of actions of B and C on A, corresponding to the induced morphisms from B and C to $[A]$.

In a *semi-abelian* category $\mathcal{C}$, an action of B on A is defined as a morphism $\xi_A^B \colon B \flat A \to A$ in $\mathcal{C}$ [1]; but a correspondence with morphisms $B \to [A]$ is only possible for *action representable* categories [2]. As a consequence, for a pair of actions (ξ_A^B, ξ_A^C) in $\mathcal{C}$, one can guarantee neither the existence nor the uniqueness of a corresponding action ξ_A^{B+C} or $\xi_A^{B \rtimes C}$.

In this talk, we will show that, in an *algebraically coherent* semi-abelian category [5], an action of $B \rtimes C$ on A is uniquely determined by a pair of actions (ξ_A^B, ξ_A^C) satisfying certain conditions. More generally, we show how *ternary semi-direct products* of A, B, C, which were previously defined by Carrasco and Cegarra in the case of groups [3] and Lie algebras [4], can be described using actions ξ_A^B, ξ_A^C, ξ_B^C and an equivariant morphism $\varphi \colon K[\xi_B^C] \to A$.

Keywords: Semi-abelian categories; actions; semi-direct products
Subject area: Categorical Algebra
MSC2020: 18E13

References

[1] F. Borceux, G. Janelidze, and G. M. Kelly. Internal object actions. *Commentationes Mathematicae Universitatis Carolinae*, 46(2):235–255, 2005.

[2] F. Borceux, G. Janelidze, and G. M. Kelly. On the representability of actions in a semi-abelian category. *Theory and Applications of Categories*, 14:244–286, 2005.

[3] P. Carrasco and A. M. Cegarra. Group-theoretic algebraic models for homotopy types. *Journal of Pure and Applied Algebra*, 75(3):195–235, 1991.

[4] P. Carrasco and A. M. Cegarra. A Dold-Kan theorem for simplicial Lie algebras. *Theory and Applications of Categories*, 32:1165–1212, 2017.

[5] A. S. Cigoli, J. R. A. Gray, and T. Van der Linden. Algebraically coherent categories. *Theory and Applications of Categories*, 30:1864–1905, 2015.